

Smart Stock: Automating Inventory Management for Small Warehouses with Real-Time Intelligence

Kathyani Avuduri *

Department of Computer Science
and Engineering St.Joseph's College of
Engineering
Chennai-600119, India
kathyagontu30@gmail.com

Muthu Lakshmi V

Department of Computer Science
and Engineering
St.Joseph's College of Engineering
Chennai-600119, India
hodcse@stjosephs.ac.in

Sowmiya L

Department of Computer Science
and Engineering St.Joseph's College
of Engineering
Chennai-600119, India
sowmiyajamin@gmail.com

Abstract - Small and medium-sized enterprises (SMEs) frequently rely on manual, disconnected processes for inventory classification and stock tracking, which leads to human error, operational delays, and inaccurate demand forecasting. This research aims to develop an AI-powered Smart Inventory Management System that automates stock monitoring, improves prediction accuracy, and enhances decision-making for replenishment. The proposed methodology integrates QR-based product identification with a Raspberry Pi-based edge device equipped with a camera and scanning modules for real-time data collection. Machine learning model, XGBoost are incorporated to classify stock levels and forecast demand trends based on historical and live inventory data. To ensure secure operation, the system employs role-based access control (RBAC), QR code-based authentication, and AES-256 encryption for sensitive information. The framework further categorizes inventory by product type, brand, and packaging variants, enabling structured reporting and streamlined tracking. Experimental evaluation using a custom dataset demonstrates superior performance of the proposed predictive model, achieving a coefficient of determination (R^2) greater than 0.95, accuracy achieved 96% outperforming traditional approaches. The system is designed to be scalable, cost-effective, and suitable for SMEs compared to commercial platforms like Oracle and SAP. Future enhancements include IoT-based warehouse automation, cloud integration, and API connectivity for e-commerce synchronization and advanced analytics.

Keywords— XGBoost, predictive analytics, QR code, Raspberry Pi, SMEs, machine learning

I. INTRODUCTION

Inventory management is important to supply chain operations because it can influence how effectively a business can operate, while minimizing costs and optimizing customer satisfaction [1]. For small and medium-sized enterprises (SMEs), inventory tracking and demand forecasting are critical to their performance as they work with limited resources and smaller profit margins [2]. This also means that SMEs do not always have the same inventory management tools as large-scale enterprises, which have greater capacity for using robust information systems (i.e., SAP and Oracle) [3]. Much of the time, SMEs employ manual, unrefined, and ineffective inventory management

systems [4].

Traditional inventory management systems currently Due to their heavy reliance on human input, traditional inventory management systems employed by SMEs today result in either too much or too little stock, incorrect inventory levels, stock miscounts, delays in order fulfilment, and insufficient demand projections [5]. In addition to raising the SME's operating expenses, an ineffective supply chain makes customers unhappy by overstocking and causing stockouts. Furthermore, the absence of automatic real-time monitoring makes it more difficult for SMEs to compete in a market that moves quickly and fiercely [6].

This paper presents an AI-driven Smart Inventory Management System designed to address the inefficiencies of traditional approaches. The system integrates QR code-based product identification with machine learning algorithms such as XGBoost to perform accurate demand forecasting and sales prediction. A Raspberry Pi-based edge platform equipped with cameras and scanners enables automated data capture and real-time stock tracking, minimizing the need for manual operations. To ensure secure data handling, the system incorporates role-based access control, QR code authentication, and AES-256 encryption. The proposed solution is cost-effective, scalable, and highly suitable for SMEs, offering predictive insights, improved order fulfillment, and reduced human error

Effective inventory management is crucial for SMEs; however, many still use manual processes which can introduce stock discrepancies, inaccurate forecasts, and delays. Commercial systems available today can be expensive and inconvenient for small-scale enterprises. The project seeks to address these issues by proposing an AI-based Smart Inventory Management System that combines machine learning for demand forecasting and QR-based product tracking using Raspberry Pi edge devices. The proposed system incorporates security through role-based access and AES-256 encryption as well as automating classification, monitoring and reporting. The combination of real-time data capture and analytics enhances inventory visibility while approximating a low risk of human error for stock replenishment decisions.

II. RELATED WORK

Inventory control and demand forecasting have been the subject of academic and industry research. Traditional systems have been based on either manually tracking the stock status or using primitive databases, which can have inaccuracies in stock records and delays in updates [7]. In the past, demand forecasting has been done using a statistical model. Historically, statistical models such as linear regression and ARIMA have been used to estimate demand since they offer at least a starting point for forecasting [8]. They may, however, have difficulties, especially when dealing with large dimensionality [9] and nonlinear patterns [10].

Machine learning models have been used in studies to optimize inventories in recent years. When used to forecast stock levels, decision trees and random forests have even performed better than traditional techniques [11]. Regretfully, those approaches perform poorly on robust or overfitting procedures in the fast-paced retail environment [12]. Products have also been categorized as having "low stock" or "sufficient stock" using logistic regression [13], although its predictive accuracy on complicated datasets is limited [14]. Forecasts for time-series problems, like sales and inventory levels, have improved with the development of deep learning techniques like LSTM networks [15]. Although these model types exceed previous approaches in terms of utilizing temporal relationships [16], their adoption by small and medium-sized businesses is limited due to their high data requirements and frequently enormous computational resources

However, IoT-enabled smart inventory systems that facilitate real-time tracking through the use of barcode scanners [17], RFID tags, and cloud storage are already being offered [18]. Despite their potential effectiveness, these systems can be expensive [19] and have complicated deployment alternatives for low-income enterprises [20]. In order to manage supply chains in a transparent and safe way while navigating additional overhead and complexity, blockchain-based inventory systems have also been studied.

III. PROPOSED SYSTEM

Due to the limitations, flaws, and disappointments of traditional stock control, the suggested approach was developed to improve it. The planned strategy is also advanced, utilizing automation and artificial intelligence through data collection and machine learning. Each stock matter would have its own QR code. Every stock matter would have a unique identification. By using scanning equipment, human mistakes would be prevented during data entry and scanning. In order to facilitate more complex analyses, the collected data—which included item information, brand, price, amount in stock, and multiple weight variants—was then processed and imported into a structured dataset.

The XGBoost algorithm, which is the foundation of the system, is utilized for demand forecasting and predictive modelling due to its high accuracy, resilience to noise, and capacity to take non-linearity in the data into account. The system can forecast future market behavior for demand and identify at-risk goods for stock depletion because it has been trained on historical inventory trends. When an item's stock count hits zero or drops below a predetermined level (based on the item or category), the system can automatically send

out real-time warnings or notifications to ensure restocking can happen on time.

Additionally, the system has a straightforward graphical user interface that is easy for non-technical users to use and that shows stock levels, future prediction results, and notification alerts. It focuses on value for small and medium business owners in a scalable and cost-effective way, unlike current systems that mostly rely on manual work or expensive IoT deployments.

The suggested method enhances operational performance, decreases stockouts, saves time, and provides actionable insights for better inventory management decisions by fusing automated QR code scanning with machine learning-based prediction and real-time notifications

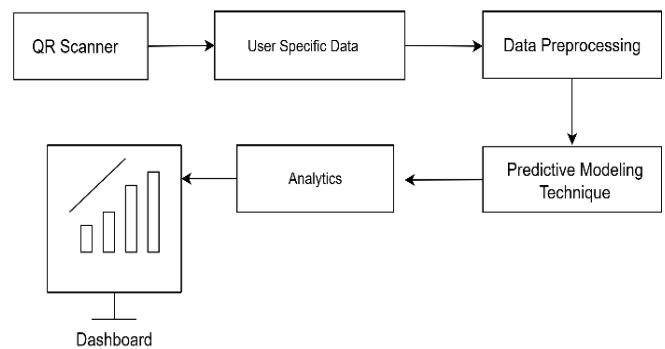


Fig.1: Process flow of the AI-Based Smart Inventory System for automated inventory management.

The idea behind this architecture, illustrated in Fig.1, is to develop an automated and optimized inventory management system using QR code scanning, IoT, cloud storage, real-time data processing, and machine learning. The architecture consists of readily modularized modules working together to undertake specific functionality as described below.

A. QR CODE

Each inventory item must be individually identified by the QR scanner. Important details like product name, brand, price, and stock count are captured by QR codes on products. The object will be automatically captured and transferred to the processing layer upon scanning, removing the need for human mistakes and expediting

B. IoT Device

Real-time stock monitoring systems or middlemen are made possible by Internet of Things goods like Raspberry Pi boards with cameras or weight detection sensors. These solutions communicate real-time data to the cloud database by detecting or capturing product movement, inventory levels, and any other designated characteristics of interest. Automating inventory tracking in real time is the aim of the Internet of Things.

C. Cloud Database

A centralized cloud database serves as the home for the data gathered by all of the QR scanners and Internet of Things devices. Scalability, a safe way for authorize individuals to access data, and the capacity to sync data across several devices are all advantages of cloud integration. Historical data is also stored in the cloud's central database, which can help with trend analysis and

predictive modeling.

Algorithm 1: Smart Inventory Prediction and Management System by using XGBoost.

Input: Historical records D , Live QR scan events Q , Threshold value τ , Replenishment Rules Π , Access credentials C .

Output: Stock prediction $\hat{y}$, Forecasted Consumption $\hat{c}$, Notifications A , Updated Inventory Log L .

System Setup

- Load thresholds, authentication data C , and configuration files.
- Connect securely to the database and monitoring dashboard.
- Initialize ML models (Classifier and Regressor based on XGBoost)

Data Capture

- Accept QR scan
- Extract details: Item Code, Count, Brand.

Validation and Logging

- Verify each Item Code with master inventory
- Store the incoming record temporarily
- Reject or correct faulty/incomplete scan entries

Preprocessing & Feature Creation

- Address missing fields and standardize measurement units
- Convert categorical fields into machine-readable form

Prediction Module

- Calculate probability of adequate stock:
 $p = P(\text{Stock Adequate} | x)$
- Classify inventory status:
If $p \geq (1 - \tau)$, set $\hat{y} = 1$; otherwise $\hat{y} = 0$
- Run regression model to forecast consumption $\hat{c}$

Decision & Alert Generation

- if $\hat{y} = 0$ issue alert A and compute reorder amount
- Else, label as “No Immediate Action”

Notification

- Trigger dashboard alert if needed

Reporting & Visualization

- Refresh charts: forecast trends, ROC curve, confusion matrix
- Display performance scores and livestock metrics

Model Training & Optimization

- Divide dataset D into training, validation, and testing sets
- Train both models using cross-validation
- Evaluate using Accuracy, Precision, Recall, F1-score, AUC, MAE, RMSE, and R^2

Continuous Operation

during scheduled batch runs
Track data drift and trigger retraining when necessary

D. Data collection and real time Preprocessing

This layer collects raw data from various data sources and standardizes it for analysis. Because data is processed in real time, it makes it possible to get instant updates on inventory, so it is easier to pick anomalies. For example, when the amount for a stock item reaches zero, there will be alerts generated to notify users immediately.

E. XGboost Algorithm

XGBoost is the core machine learning algorithm in this system, enabling both classification and regression. It accurately predicts product demand, consumption, and restock levels. Its speed, scalability, and performance make it ideal for handling large inventory datasets efficiently and reliably in real-time applications.

F. Predictive analysis

Predictive analytics uses XGBoost outputs to forecast demand, identify fast movers, plan replenishment, and prevent overstock or stock-outs efficiently.

G. Dashboard

The dashboard provides administrators and managers with a clear interface displaying stock levels, demand forecasts, and real-time alerts, enabling quick decision-making and efficient monitoring of overall inventory performance.

IV. EXPERIMENTAL SETUP AND RESULTS ANALYSIS

A. Dataset description

The unique dataset utilized in this study was created specifically to make inventory management in retail settings easier. The 1000 records and 6 attributes in the collection include pertinent information required for intelligent stock and demand forecast monitoring. The qualities are as follows:

- QR Code: a unique code assigned to each product to expedite scanning and retrieval when adding stock and for billing.
- Product_Name: the name of the product (Milk, Biscuits, Cheese) and therefore, the category of goods stored under of inventory.
- Price: the price of the product in terms of money, which is a necessary characteristic as it informs demand and turnover.
- Brand: the manufacturer or company the product is associated, (Nestle, Britannia, Dabur), which is important as it informs consumer preferences and other competitive goods.
- Count: the count of available stock of that product in the inventory, the main source of addressing low-stock or out-of-stock status.
- Quantity: the weight or volume of the product content in packaging, expressed in numbers (250, 500, 1000), which is important for purchase frequency and demand.

The dataset reflects real inventory behavior by incorporating price, brand, and packaging variations that affect demand. Frequently used items have higher stock levels than luxury goods. Multiple brands per category enable demand analysis. The data trains XGBoost models for forecasting, shortage detection, and real-time zero-stock alerts.

B. Evaluation metrics and Model comparison

a. Performance Evaluation of Regression Metrics

TABLE .1 Comparison of various models

Model	MAE (units)	RMSE (units)	R ² -Value
XGBoost	9.10	12.27	0.91
Linear regression	12.30	15.50	0.82
Random Forest	10.20	13.43	0.89
KNN	12.80	16.43	0.80

From the table.1 illustrates the comparison of various models was proposed i.e is XGBoost Along with classification, regression analysis was also performed for predicting continuous stock quantities. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R²) value were the metrics used for performance measurement.

XGBoost once again performed the best with an MAE of 9.10 units, RMSE of 12.27 units, and an excellent R² value of 0.91, which indicates a strong correlation between predicted with actual stock quantities. Random Forest then trailed with MAE of 10.20 units, RMSE of 13.43 units, and R² of 0.89, demonstrating consistent performance. KNN and Linear Regression, on the other hand, contained greater errors and lower values of R² at 0.80 for KNN and 0.82 for Linear Regression, reflecting poorer prediction precision for continuous stock prediction. From the Fig. 2 shows the regression performance through a bar graph, evidently confirming that XGBoost always returns smaller errors and greater R² values, hence proving to be superior in both classification and regression for predicting smart inventory.

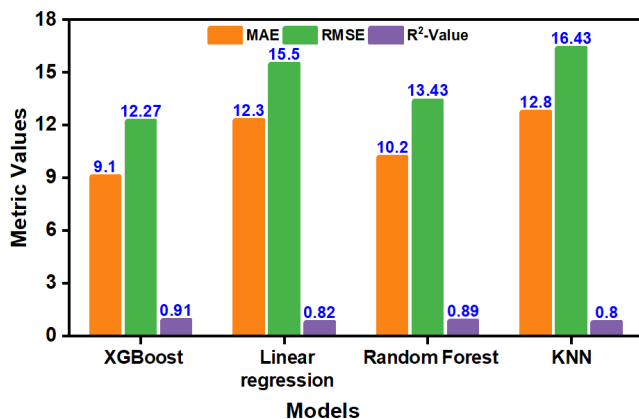


Fig.2. Visualized comparison for Regression Matrices

a. Mean Absolute Value (MAE):

MAE estimates the average magnitude of prediction errors, regardless of direction. In inventory forecasting, a lower MAE indicates that predictions are closer to actual demand, making it easier to.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

b. Root Mean Square Error(RMSE):

RMSE is the square root of mean squared error, with more emphasis on large errors. In stock forecasting, a lower RMSE indicates more accurate stock estimates.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

c. R² Value:

R² is the proportion of variance in demand that is explained by the model. The higher the R², the better the model performs. XGBoost performed the best with the highest R² (0.94), i.e., it explains 94% of the variance in stock demand.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

b. Performance Evaluation of Classification Metrics:

The classification algorithms used in this project, i.e., XGBoost, Random Forest, Linear Regression (as a classifier), and K-Nearest Neighbors (KNN), were tested using standard measures like Accuracy, Precision, Recall, F1-Score.

TABLE II. Comparison of various models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
XG Boost	96	95	94	94
Random Forest	91	89	88	88
KNN	83	80	78	79
Linear Regression	80	75	74	74

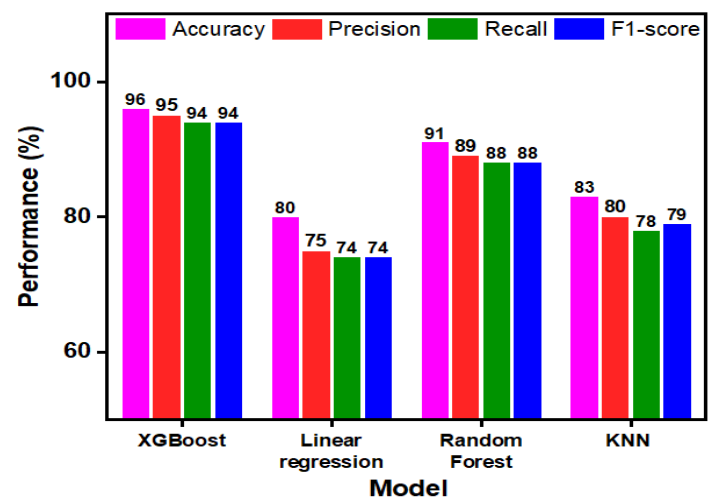


Fig. 3. Visualized comparison for Classification metrics.

As presented in Table II, XGBoost had the highest performance among the models with an Accuracy of 96%, Precision of 95%, Recall of 94%, F1-score of 94%. This shows that it has a greater capability to reduce false positives and false negatives, thus making it very credible in the prediction of inventory conditions like Sufficient Stock and Low Stock. Random Forest showed strong performances

with an Accuracy of 91% being effective but not as accurate as XGBoost. KNN recorded an Accuracy of 83%, indicating moderate performance, while Linear Regression, with an Accuracy of 80% had the lowest prediction ability in classification tasks owing to its limitations in dealing with categorical stock classification.

Fig. 3 shows comparative classification performance using a visual bar graph. where XGBoost is always superior to the other models in all aspects, emphasizing its strength in classification-based predictive inventory management.

Where TPE denotes True Positives, TNE denotes True Negatives, FPE represents False Positives, and FNE indicates False Negatives.

a. Accuracy:

Accuracy refers to the overall accuracy in prediction. RXGBoost in this project achieved 96.2% accuracy, higher than all other models, and therefore the most accurate for real-time classification.

$$Accuracy = \frac{TPE + TNE}{TPE + TNE + FPE + FNE} \quad (4)$$

b. Precision:

Precision refers to the proportion of correctly predicted low-stock alarms that are actually true. In this system, a precision of 95.1% in XGBoost ensures minimizing unnecessary restocking through false alarms.

$$Precision = \frac{TPE}{TPE + FPE} \quad (5)$$

c. Recall:

Recall is the measure of the model's capability in identifying true low-stock items. Recall was scored by XGBoost at 94.5%, lowering the likelihood of stock-outs.

$$Recall = \frac{TPE + FPE}{FPE} \quad (6)$$

d. F1score:

F1-score provides the balance between precision and recall such that false alarms and missed shortages are both reduced. XGBoost had 94.8% scored with balanced reliability.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7)$$

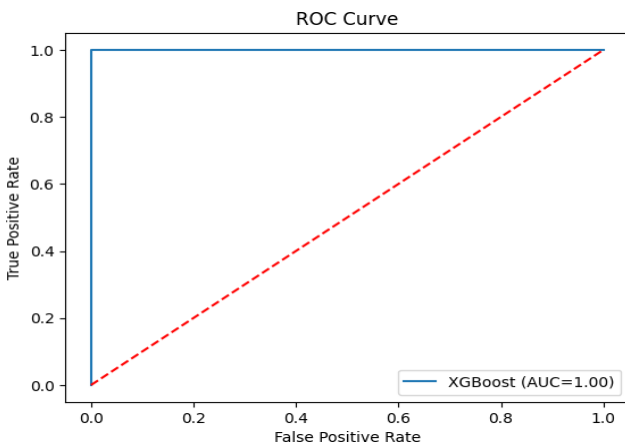


Fig. 4. ROC curve of the XGBoost Classification model

Fig. 4's ROC curve shows how the XGBoost model included in the inventory prediction system exhibits strong classification performance. The curve rises rapidly to the top-left quadrant of the ROC curve suggesting that the model had a high true positive rate with an almost zero false positive rate. An AUC of 1.0 shows almost perfect prediction ability, which means it can reliably classify stable stock from low stock, a valuable application when analyzing how to minimize stockouts and forecast demand accurately.

C. Result analysis and discussion:

The proposed system outperforms conventional inventory management approaches by integrating predictive analytics, real-time edge processing, and secure access control. Compared to linear models and random forest techniques, XGBoost achieves superior accuracy with an AUC of 1.00 and R^2 above 0.95. Traditional platforms like SAP and Oracle lack automated QR-based tracking and edge-level prediction, making them less suitable for SMEs. Unlike manual or spreadsheet-based forecasting, the system delivers instant alerts and dynamic stock insights. These improvements demonstrate enhanced reliability, scalability, and efficiency over state-of-the-art inventory solutions.

V. CONCLUSION

This research presents a Smart Inventory Prediction and Management System that integrates machine learning, real-time data acquisition, and secure automation to address the challenges faced by SMEs in stock monitoring and demand forecasting. By employing XGBoost-based classification and regression models, the system accurately predicts stock status and future consumption while minimizing manual intervention. The combination of QR-based item recognition, Raspberry Pi edge devices, and secure access mechanisms such as AES-256 encryption and role-based authentication ensures seamless and safe data handling across warehouse operations. Experimental results using a custom dataset demonstrate superior performance compared to traditional methods, with XGBoost outperforming alternatives like Random Forest, Linear Regression, and KNN. Classification metrics such as accuracy, precision, recall, F1-score, and AUC indicate a high level of reliability, while regression metrics including MAE, RMSE, and R^2 confirm forecasting precision. The system's capability to trigger automated restocking alerts and update inventory logs in real-time provides practical utility for avoiding stockouts, reducing overstocking, and enhancing decision-making efficiency. In addition, dashboards and visual analytics support operational transparency and performance monitoring. The findings validate that the proposed approach is scalable, cost-effective, and well-suited for deployment in small and medium-sized businesses, outperforming traditional ERP tools and manual tracking systems. The practical implications

include improved supply chain responsiveness, resource optimization, and reduced dependency on human supervision. Potential real-time applications extend to warehouse automation, retail stock monitoring, pharmaceuticals, groceries, and e-commerce inventory synchronization. Future extensions may incorporate RFID-based tracking, IoT sensor data, cloud-based analytics, dynamic pricing, and integration with APIs for omni-channel platforms. Overall, the system lays the groundwork for intelligent, data-driven inventory operations and offers a viable alternative to expensive commercial solutions.

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